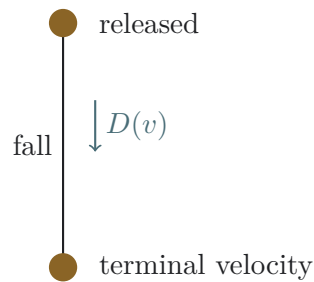


Is Drag One Law, or Many?

Name: _____

A falling object speeds up until, eventually, it stops speeding up, and the law behind how it got there turns out to depend on how fast it was already going.



The question. Something pushes back on a falling object exactly as hard as gravity pulls, once it stops accelerating. How does that push depend on how fast the object is already moving?

Working rule: No exponent for the speed-dependence is assumed, and no promise is made that the same law applies at every speed.

1. Something pushes back

the balance condition At

terminal velocity, the forces balance.

$$mg = D(v_t)$$

D is deliberately left unspecified. State two properties D must have, just from this equation and the physical setup.

2. Two candidate laws

linear or quadratic

$$D(v) = k_1 v$$

$$D(v) = k_2 v^2$$

For each, describe the physical picture it suggests (a thick fluid dragging smoothly, or air being shoved aside).

3. Testing by weight, not by speed

an indirect test Speed

cannot be dialled in directly, since the object settles on it by itself. Weight can, by stacking identical objects. Using $D(v) = kv^n$,

$$mg = kv_t^n \quad \implies \quad v_t \propto m^{1/n}.$$

Complete the two predictions.

If $n = 1$ doubling the weight should _____

If $n = 2$ doubling the weight should _____

4. Take real measurements

terminal speed against weight Us-

ing stacked coffee filters or a similar object, record terminal speed at several weights (or use the values given).

Number stacked, n	Terminal speed, v_t	v_t/n^1	$v_t/n^{1/2}$
1	0.85		
2	1.15		
4	1.55		
8	2.05		

Which column stays closer to constant? What does that suggest about n at this range of weights?

5. The crucial complication a balloon is not a point charge, and a filter is not a point mass List

two ways a real falling object departs from the idealised model.

1. _____

2. _____

6. The regime that decides the Reynolds number

$$Re = \frac{\rho v L}{\eta}$$

State, in words, what each symbol represents, and which regime (small Re or large Re) favours each candidate law.

Small Re _____

Large Re _____

7. Not one law, but a family the unified picture

$$D(v) = k_1 v + k_2 v^2$$

Explain why this single expression behaves like each pure candidate in its own regime, without needing to choose between them in advance.

8. From force to potential

revisit an earlier idea Re-

call that force and potential energy are related by $F = -dU/dr$. Explain, in one or two sentences, why this relation is harder to apply usefully to drag than to Coulomb's law.

9. Interaction or constitutive

classify drag

Coulomb, Lorentz	interaction law, fixed exponent
Drag	

Drag is velocity-dependent, which sits it beside the Lorentz force by one axis, yet it belongs beside Ohm's law by another. Explain the mismatch in your own words.

10. Core results

what this enquiry actually earned The

relations established along the way.

$$mg = D(v_t)$$

$$D(v) = k_1 v + k_2 v^2$$

$$v_t \propto m^{1/n}$$

$$Re = \frac{\rho v L}{\eta}$$

Concepts earned, in order.

- Terminal velocity is a force balance, not a speed limit imposed from outside.
- Weight, not speed, is the variable that can actually be controlled directly.
- The fitted exponent is allowed to drift with the range of weights tested, and that drift is real physics, not error.
- A single two-term law explains why two seemingly rival candidates both work, each in its own regime.
- Drag is velocity-dependent but constitutive, a genuine mismatch between the two organising questions of this series.

Every substantial claim should be accompanied by a reason, calculation, table, or counterexample.